shows the structure of the ...

BCNF OR 3.5 Normal form.
(Boyce-Codd Normal form)

→ Table should satisfy 2 conditions.

- it should be in the 3rd Normal form.

- for any dependency A→B, A should be a Superkey.

which means.

for, A → B

If A is non-prime and B is a prime attribute

In 3rd NF We Learn

Non prim Attb → non prime attr

Transitive Dep.

• scopes and rules are followed.

→ But What if

A non-prime attribute → (deriving) prime Attribute

BCNF doesn't allow this

ex: college Enrollment Table.

st-id	subj	prof.
101	java	p. java
101	c++	p. cpp
102	java	p. java22
103	c#	p. chash
104	java	p. java

→ as we can see one student can enroll in multiple subjects and for each subject professor is assign to the student.

Note: there is multiple professors teaching one subj.

Like we have jane.

~ so here st-id and subject are primary key.

~ so professor → can lead subject. and subject is the part of candidate key primary key hence it is the prime attribute (subject).

while professor is the non-prime attribute.

hence we have a dependency here. that ~ subject is depend on professor. but professor is not a superkey.

~ so the table doesn't satisfy BCNF.

~ so make the table in BCNF :-

College Enrollment Table

↓

~ Student Table + professor Table

ex:-

~ Student Table.		Professor Table.		
~ st_id	p-id	P-id	professor	subject.

Now our tables satisfy BCNF.